# Monday.com Integration

## 1. Overview

### 1.1 Description

**Monday.com Integration** connects **Monday.com** (work management platform) with **Newo Platform**.

It enables:

- Checking appointment availability based on existing meetings in Monday boards
- Creating new bookings as meeting items in Activities board with contact linking
- Cancelling appointments by archiving meeting items
- Detecting existing customers by phone and retrieving their upcoming bookings
- Managing two interconnected boards: Activities (meetings) and Contacts (customers)
- Optional silent availability preload on conversation start

This integration is designed for **service businesses and teams** who use Monday.com boards for scheduling and need **automated appointment management through a conversational AI agent** (voice or chat).

### 1.2 Use Cases

- When a client asks about availability → Query Activities board for meetings, compute free slots
- When a client wants to book → Find or create contact in Contacts board, create meeting in Activities board
- When a client wants to cancel → Archive the meeting item in Activities board
- When a caller's phone is identified → Look up contact and retrieve upcoming bookings
- When a conversation starts → Optionally preload nearest available slots silently

### 1.3 Supported Features

| Feature | Supported | Notes |
|---|---|---|
| Check Availability | ✅ | Computes free slots from booked meetings in Activities board |
| Book Appointment | ✅ | Creates meeting item with contact link |
| Cancel Appointment | ✅ | Archives meeting item |
| Existing Client Detection | ✅ | By phone number, with prompt context injection |
| Upcoming Booking Retrieval | ✅ | Future meetings stored in persona |
| Silent Availability Preload | ✅ | Optional check on conversation start |
| Contact Auto-Creation | ✅ | If contact not found by phone, creates new one |
| Custom Date/Notes Columns | ✅ | Auto-detected from board structure |
| Configurable Slot Duration | ✅ | Default 30 min |
| Reschedule | ❌ | Cancel + rebook as workaround |

| | | |
|---|---|---|
| Board Template Setup | ❌ | Boards must be pre-created manually |

## 2. Requirements

Before installation:

- Active **Monday.com** account
- **Monday.com OAuth App** with Client ID and Client Secret
- Two pre-configured boards:
  - **Activities board** — for meetings/events (must have columns: `activity_type`, `activity_status`, `activity_start_time`, `activity_end_time`, `activity_item`)
  - **Contacts board** — for customer records (must have column: `contact_phone`)

## 3. How to Setup

### 3.1 Step 1 — Authorize in Monday.com

1. Create an OAuth app in Monday.com developer portal
2. In Newo platform, set `monday_client_id` and `monday_client_secret`
3. Obtain the OAuth authorization code
4. Paste the code into the `monday_oauth_code` attribute

### 3.2 Step 2 — Connect in Platform

1. Open **Newo → Projects**
2. Set the following attributes in **Builder / Attributes**:

| Attribute | Required | Description |
|---|---|---|
| `monday_client_id` | ✅ | OAuth Client ID |
| `monday_client_secret` | ✅ | OAuth Client Secret |
| `monday_oauth_code` | ✅ | One-time OAuth authorization code |
| `activities_id` | ✅ | Board ID for meetings/activities |
| `contacts_id` | ✅ | Board ID for contacts |
| `monday_base_api_url` | ❌ | GraphQL API endpoint (default: `https://api.monday.com/v2`) |
| `monday_timezone` | ❌ | Booking timezone (default: project timezone) |
| `monday_duration` | ❌ | Slot duration in minutes (default: `30`) |
| `monday_show_for_days` | ❌ | Days ahead for availability (default: `1`) |
| `monday_check_availability_on_conversation_start` | ❌ | Auto-check availability on start (default: `False`) |
| `monday_slots_available` | ❌ | Enable availability checking (default: `True`) |

| | | |
|---|---|---|
| `monday_booking_available` | ❌ | Enable booking capability (default: `True`) |
| `monday_cancelation_available` | ❌ | Enable cancellation capability (default: `True`) |

3. Click **Save + Publish All**
4. On publish, the SetupFlow automatically:
    - Exchanges the OAuth code for an access token
    - Queries the Activities board for column structure (auto-detects date and notes columns)
    - Discovers the group ID for activity items
    - Injects booking and availability payload schemas

# 4. How to Use

## 4.1 How the Integration Works

The integration uses Monday.com **GraphQL API** to manage two boards: Activities (meetings) and Contacts (customers). Availability is computed by querying existing meetings and finding gaps. Bookings create new items linked to contacts.

- A **Trigger** is a system event that starts a flow
- An **Action** is the GraphQL operation performed in Monday.com

**Available Triggers**

| Trigger | Event | Description |
|---|---|---|
| **Check Availability** | `get_available_slots` | When the agent needs to look up open slots |
| **Book Appointment** | `book_slot` | When the agent confirms a booking |
| **Cancel Appointment** | `convo_agent_cancel_booking` | When the agent processes a cancellation |
| **Phone Identified** | `define_user_phone_number` | When the caller's phone is detected |
| **Session Start** | `session_started` | Optional silent availability preload |
| **Setup** | `start_update_event` | Initializes integration on deploy |

**Available Actions**

- **Query Activities** — Fetch meetings with filters (GraphQL `items_page` on Activities board)
- **Create Meeting** — New meeting item with dates and contact link (GraphQL `create_item`)
- **Archive Meeting** — Cancel appointment (GraphQL `archive_item`)
- **Query Contacts** — Search by phone (GraphQL `items_page` on Contacts board)
- **Create Contact** — New contact with phone (GraphQL `create_item`)

## 4.2 Architecture Diagrams

**Overall Sequence**

```
sequenceDiagram
    participant U as User
    participant A as ConvoAgent
```

```
    participant I as Monday Integration
    participant API as Monday.com GraphQL

    Note over I,API: Setup (on publish)
    A->>I: start_update_event
    I->>API: POST /oauth2/token (code)
    API-->>I: access_token
    I->>API: Query board columns + groups
    API-->>I: Column structure, group ID

    Note over U,API: Existing Client Detection
    A->>I: define_user_phone_number (phone)
    I->>API: Query Contacts board (phone filter)
    API-->>I: Contact data
    I->>A: Inject ExistingCustomerInfo into prompt
    I->>API: Query Activities board (contact_id link)
    API-->>I: Future meetings
    I->>A: Store bookings in persona

    Note over U,API: Availability Check
    U->>A: "Any slots tomorrow?"
    A->>I: get_available_slots (date, time)
    I->>API: Query Activities (type=Meeting, status≠Done, date range)
    API-->>I: Booked meetings
    I->>I: Compute free slots (subtract booked, slice by duration)
    I->>A: result_success (availability[])
    A->>U: "Here are the available times..."

    Note over U,API: Booking
    U->>A: "Book 2:00 PM"
    A->>I: book_slot (customerInfo, dateTime)
    I->>API: Query Contacts (phone)
    alt Contact not found
        I->>API: Create Contact (name, phone)
        API-->>I: contact_id
    end
    I->>API: Create Activity (Meeting, start/end, contact link)
    API-->>I: item_id
    I->>A: result_success (booking_id)
    A->>U: "Your appointment is confirmed!"

    Note over U,API: Cancellation
    U->>A: "Cancel my appointment"
    A->>I: convo_agent_cancel_booking (booking_id)
    I->>API: Archive item (booking_id)
    API-->>I: Confirmation
    I->>A: result_success
    A->>U: "Cancelled"
```

**AvailabilityFlow**

```
flowchart TD
    E["get_available_slots<br>or session_started"] --> GATE{{"Slot
check<br>enabled?"}}
    GATE -->|"No"| SKIP["Skip"]
    GATE -->|"Yes"| QUERY["GraphQL query:<br>Activities
board<br>type=Meeting<br>status≠Done<br>date range filter"]
    QUERY --> PARSE["Extract booked events:<br>start_time, end_time"]
    PARSE --> CALC["Sliding window:<br>generate slots at<br>duration
intervals<br>skip overlapping"]
    CALC --> FORMAT["Group by date<br>Format as HH:MM AM/PM"]
    FORMAT --> OUT["result_success<br>{availability[]}"]
```

**BookingFlow**

```
flowchart TD
    E["book_slot"] --> GATE{{"Booking<br>enabled?"}}
    GATE -->|"No"| SKIP["Skip"]
    GATE -->|"Yes"| CONTACT["Query Contacts<br>board by phone"]
    CONTACT --> FOUND{{"Contact<br>found?"}}
    FOUND -->|"No"| CREATE["Create Contact:<br>name + phone"]
    CREATE --> CID["contact_id"]
    FOUND -->|"Yes"| CID
    CID --> UTC["Convert datetime<br>to UTC"]
    UTC --> ITEM["Create Activity item:<br>'Meeting with {name}'<br>type=Meeting,
status=Open<br>start/end times<br>link to contact"]
    ITEM --> OUT["result_success<br>{booking_id}"]
```

**CancellationFlow**

```
flowchart TD
    E["convo_agent_cancel_booking"] --> GATE{{"Cancellation<br>enabled?"}}
    GATE -->|"No"| SKIP["Skip"]
    GATE -->|"Yes"| ID["Extract booking_id<br>from payload.meta"]
    ID --> ARCHIVE["GraphQL mutation:<br>archive_item<br>{item_id: booking_id}"]
    ARCHIVE --> OUT["result_success"]
```

**ExistingClientFlow**

```
flowchart TD
    E["define_user_phone_number"] --> SEARCH["Query Contacts board<br>{phone
filter}"]
    SEARCH --> FOUND{{"Contact<br>found?"}}
    FOUND -->|"No"| DONE["No action"]
    FOUND -->|"Yes"| INJECT["Inject ExistingCustomerInfo<br>into prompt context"]
    INJECT --> APPTS["Query Activities board<br>{contact_id link,<br>status=Open}"]
    APPTS --> FILTER["Filter future<br>meetings only"]
    FILTER --> SAVE["Store in persona<br>attribute: bookings"]
```

### 4.3 Where to Test

Testing can be done through the **Newo conversation interface** (chat or voice channel).

### 4.4 Testing and Verification

**To test the integration:**

1. **Setup**: Publish and verify OAuth token + board structure discovery
2. **Existing Client**: Call from a known phone — verify customer info injected into prompt
3. **Availability**: Ask about open slots — verify free times computed from Activities board
4. **Booking**: Complete a booking — verify meeting item created in Activities board linked to contact
5. **Cancellation**: Cancel a booking — verify item archived in Monday.com

**If no action occurs:**

- Ensure OAuth credentials are set ( `monday_client_id` , `monday_client_secret` )
- Verify `monday_access_token` was obtained
- Confirm `activities_id` and `contacts_id` are set to valid board IDs
- Check that the Activities board has required columns ( `activity_type` , `activity_status` , `activity_start_time` , `activity_end_time` )
- Verify feature flags are enabled
- Re-publish the project if settings were changed

> *Note: This integration creates and archives real items in Monday.com boards.*

## 5. FAQ

**Q: What board structure is required?** A: Two boards are needed. The **Activities board** must have columns: `activity_type` (Meeting), `activity_status` (Open/Done), `activity_start_time` , `activity_end_time` , `activity_item` (relationship to Contacts). The **Contacts board** must have `contact_phone` column.

**Q: How is availability computed?** A: The integration queries the Activities board for meetings (type=Meeting, status≠Done) within the date range, then uses a sliding window algorithm to find gaps. Slot duration is configurable via `monday_duration` (default: 30 min).

**Q: How does cancellation work?** A: Cancellation uses Monday.com's `archive_item` mutation — it doesn't delete the item but archives it, making it recoverable. The booking_id is the Activities board item ID.

**Q: Are date and notes columns auto-detected?** A: Yes. During setup, the integration queries the Activities board columns and auto-detects date columns (by `date_` ID prefix or "Date" title) and notes columns (by `long_text_` ID prefix or "Notes" title). If not found, items are still created without these optional columns.

**Q: Can I use existing Monday.com boards?** A: Yes, as long as they have the required column structure. Set the board IDs in `activities_id` and `contacts_id` attributes.

**Q: What happens with token refresh?** A: The UtilsFlow handles HTTP 401 responses, but the token refresh implementation for Monday.com is currently a placeholder. If tokens expire, re-authorization may be needed.